RENSHAW DRIVE ENGINEERING CONSULTANTS

Independent Expert Report

Ashfield Dynamics Limited v Marwick Drives Limited — Claim No. BL-2025-000412

Report of Dr Alan Renshaw CEng FIMechE on the cause of failure of MD-450 gearmotor units

Dated 12 September 2025

1. Instructions and expertise

I am instructed by Whitfield Carter LLP, solicitors for the Claimant, to investigate and report on the cause of the premature failures of MD-450 helical-bevel gearmotor units supplied by the Defendant. I am a Chartered Engineer and Fellow of the Institution of Mechanical Engineers with 28 years' experience in the design and failure analysis of industrial power transmission equipment.

I understand my overriding duty is to the Court and not to those instructing me, and I have complied with that duty. This report sets out my independent opinion.

2. Documents and samples examined

I examined six failed units (serial numbers MD450-0117, 0124, 0131, 0146, 0152 and 0159), the installation and laser-alignment records, the planned-maintenance logs, the line duty data downloaded from the control system, and the Defendant's installation and maintenance manuals.

3. The failure mode

All six units exhibited the same failure mode: collapse of the output-shaft bearing followed by loss of gear-mesh geometry and stripping of the bevel pinion teeth. Spectrographic and hardness testing of the pinions showed an effective case depth of 0.3–0.4 mm against a design requirement of 0.8–1.0 mm, indicating that the components had been under-carburised during heat treatment.

Bearing dimensional checks showed that the fitted output bearing (a single-row 6309 type) is one size below that required for the rated radial load of the MD-450 at continuous duty. The calculated L10 bearing life at the units' rated load is approximately 5,800 hours, which corresponds closely to the in-service failures observed at six to nine months of continuous operation.

4. The Defendant's alternative explanations

I considered the Defendant's contention that misalignment, overloading and poor maintenance caused the failures. The alignment records show installation within the manufacturer's tolerance of 0.10 mm; the recorded line loads are within the units' nameplate rating; and the maintenance logs show lubrication carried out to schedule. None of these explanations is consistent with the metallurgical evidence, which points unambiguously to an inherent manufacturing deficiency.

5. Conclusions

In my opinion, on the balance of probabilities, the failures of the MD-450 units were caused by (i) an undersized output bearing and (ii) inadequate case-hardening of the bevel pinion. Both are defects of design and manufacture for which the Claimant bears no responsibility. The failure rate of 41 in 60 units within the warranty period is consistent with a systemic manufacturing defect rather than misuse.

Statement of Truth

I confirm that I have made clear which facts and matters referred to in this report are within my own knowledge and which are not. I believe that the facts I have stated in this report are true and that the opinions I have expressed are correct. I understand my duty to the Court and have complied with it.

Dr Alan Renshaw CEng FIMechE

Renshaw Drive Engineering Consultants — 12 September 2025